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Fourier-enhanced high-order total variation (FeHOT) iterative network for interior tomography

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Abstract

Objective. Determining a satisfactory solution for different computed tomography (CT) fields has been a long-standing challenge in the interior tomography, Traditional methods like FBP suffer from low contrast, while deep learning approaches often lack data consistency. The goal is to leverage high-order total variation (HOT) regularization and Fourier-based frequency domain enhancement to achieve high-precision reconstruction from truncated projection data while overcoming limitations such as slow convergence, over-smoothing, and loss of high-frequency details in existing methods. *Approach*. The proposed Fourier-enhanced HOT (FeHOT) network employs a coarse-to-fine strategy. First, a HOT-based unrolled iterative network accelerates coarse reconstruction using a learned primal-dual algorithm for data consistency and implicit high-order gradient constraints. Second, a Fourier-enhanced U-Net module selectively attenuates low-frequency components in skip connections while amplifying high-frequency features from filtered back-projection (FBP) results, preserving edge and texture details. Frequency-dependent scaling factors are introduced to balance spectral components during refinement. *Main Results*. Experiments on the AAPM and clinical medical datasets demonstrate FeHOT's superiority over competing methods (FBP, HOT, AG-Net, PD-Net). For the medical dataset, FeHOT achieved PSNR = 41.17 (noise-free) and 39.24 (noisy), outperforming PD-Net (33.42/31.08) and AG-Net (33.41/31.31). Meanwhile, For the AAPM dataset, where imaged objects exhibit piecewise constant properties, first-order total variation achieved satisfactory results. In contrast, for clinical medical datasets with non-piecewise-constant characteristics (e.g. complex anatomical structures), FeHOT's second-order regularization better aligned with the high-quality requirements of interior tomography. Ablation studies confirmed the necessity of Fourier enhancement, showing significant improvements in edge preservation (e.g. SSIM increased from 0.9877 to 0.9976 for noise-free cases). The method achieved high-quality reconstruction within five iterations, reducing computational costs. *Significance*. FeHOT represents a paradigm shift in interior tomography by: 1) Bridging classical HOT theory with deep learning through an iterative unrolling framework. 2) Introducing frequency-domain operations to overcome the limitations of polynomial/piecewise-constant assumptions in CT images. 3) Enabling high-quality reconstruction in just five iterations, balancing computational efficiency with accuracy. This method offers a promising solution for low-dose, precise imaging in clinical and industrial applications.

1. Introduction

In the field of imaging, particularly x-ray computed tomography (CT), the potential hazards posed by exposure to ionizing radiation remain a long-standing concern. Traditionally, the approach to reconstruction involves scanning the entire object; however, only a fraction of the scanned object is analyzed. An effective

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method involves performing local scanning of the region-of-interest (ROI) and reconstructing the ROI from the obtained truncated projections, also known as interior tomography. ROI reconstruction offers an effective strategy for attaining high-precision reconstruction while prioritizing radiation protection and adhering to the principle of minimizing exposure to levels that are as low as reasonably achievable.

Interior tomography constitutes an ill-posed inverse problem. Truncated data result in low contrast in the reconstructed images using conventional algorithms such as filtered back projection (FBP) (Feldkamp *et al* 1984, Quinto 1988). For ROIs not entirely within scanned objects, in 2004, Zou and Pan (2004), Pan *et al* (2005) proposed a BPF algorithm based on PI-lines (which is defined to be any line that has two points on the helix less than 360 degrees apart (Danielsson *et al* 1997). This algorithm reconstructs images through differentiation, back projection, and filtering (finite Hilbert transform), demonstrating efficiency in handling data truncation. However, this method is noticeably sensitive to noise, due to the differentiation of projection data, resulting in inferior reconstructed image quality in practice.

When the ROI is completely located inside a scanned object, interior tomography does not have any unique solution (Natterer 2001). However, certain research results have shown that, under the premise of providing some prior knowledge, the internal ROI can attain unique and stable solutions. This is an important breakthrough in ROI imaging research (Courdurier *et al* 2008, Kudo *et al* 2008, Yu *et al* 2009b). Consequently, many researchers have proposed model-based iterative algorithms (MBIR) for solving such internal problems. These iterative algorithms leverage prior image assumptions, and the reconstructed image has certain properties. For example, compressed sensing (CS) theory assumes that the image has sparsity and subsequently achieves signal recovery through an ℓ_1 norm constraint (Candès *et al* 2006, Donoho 2006, Yu and Wang 2009, Yu *et al* 2009a). Simultaneously, TV regularization methods have been employed in many studies to handle these reconstruction problems (Ziegler *et al* 2008, Hamelin *et al* 2010, Paleo and Mirone 2017). Although these methods enable high-contrast imaging, they can only effectively reconstruct images approximated well by piecewise constant functions. In order to break through this constraint, Yang *et al* proposed a high-order total variation (HOT) algorithm to accurately reconstruct piecewise-linear or polynomial functions (Yang *et al* 2011, 2012a, 2012b, Zhao *et al* 2015). Although these studies have achieved highly accurate reconstruction images compared with analytical reconstruction methods, MBIR converges slowly and involves forward and back projection operators during the iteration process, resulting in significant time costs. Moreover, clinical ROIs often violate prior assumptions, leading to over-smoothed results or laborious parameter tuning.

In recent years, deep learning has emerged as a promising alternative, leveraging powerful learning capabilities and rapid inference. In the field of interior tomography, Han and Ye (2019) have proposed using DBP reconstruction images instead of FBP reconstruction images as the input of the U-Net for better stability in various types of ROI reconstruction tasks (Huang *et al* 2021). Additionally, interior tomography images can be directly learned from truncated data by the iCT-Net (Li *et al* 2019, Yinsheng *et al* 2019). However, postprocessing CNNs lack theoretical guarantees and generalizability, failing to ensure data consistency (Fawzi *et al* 2017, Liu *et al* 2018, Yuan *et al* 2019). To ensure the robustness of reconstructed images and the consistency of data, many researchers have proposed various iterative models to achieve robust reconstruction for CT problems (Adler and Öktem 2018, Zhang and Ghanem 2018, Bertocchi *et al* 2020, Monga *et al* 2021, Emmanuel Komolafe *et al* 2022, 2024). Recent studies have shown that using deep learning without considering the impact of high-frequency information on the network can easily lead to overlooking details. For example, FreeU (Si *et al* 2024, Wang *et al* 2024) finds that the main backbone of U-Net is critical for denoising, whereas its skip connections primarily contribute high-frequency features to the decoder. Concatenating these two features directly leads to the network ignoring the semantic content of the primary feature. Therefore, reducing the low-frequency components of the skip feature and enhancing the primary feature also necessary.

We propose an iterative network combining high-order regularization in the image domain with frequency-domain weighting to recover details and transcend piecewise-constant/polynomial assumptions. First, we unroll the HOT framework into a high-order TV network, enabling efficient iterative reconstruction. Second, we integrate Fourier transforms into U-Net skip connections to preserve FBP-derived high-frequency details, countering HOT's structural assumptions while mitigating cupping artifacts and CT value shifts. This dual-domain approach achieves precise special solutions with verified reproducibility. Our code is available at https://github.com/GM0322/FeHOT.

The key contributions of this paper can be summarized as follows:

- We propose an iterative network based on the high-order TV method for reconstructing ROI problems. This network can effectively improves the speed of reconstruction iterations, achieving high-accuracy reconstructions in as few as five iterations.

- We employ a Fourier transform in the skip connections of the U-Net model to introduce high-frequency features from each iteration into the decoder through frequency domain processing. This approach effectively fuses high-frequency information from FBP-reconstructed images, achieving significant detail improvement.
- A novel reconstruction strategy facilitates the reconstruction process, employing a coarse-to-fine approach. Coarse reconstruction is initially achieved using the high-order TV theory, followed by detail supplementation through high-frequency feature introduction. This method ultimately achieves high-precision fine reconstruction.
- We introduce frequency features reweighting in the backbone of the work model. This novel operation refreshes the connection or communication modes between network layers. Here, we demonstrate its advantage in restoring high-frequency details. which also may have other potential special effects in other fields.
- This process represents a transformation from one particular solution to another, providing a method to discover a new particular solution via feature introduction.

The remainder of this paper is organized as follows: section 2 reviews the HOT for the interior tomography. Section 3 presents the proposed method and its application to the interior tomography. The proposed method is evaluated through an ablation study in section 4. Related issues are discussed and conclusions are drawn in section 5.

2. Review of the HOT for interior tomography

In a general case, consider a scanned object $f(x_1,x_2)$ that is compactly supported on $\mathbb{R}^2$. The Radon transform, denoted Rf, is defined on the space of straight lines $L \subset \mathbb{R}^2$.

$$Rf(L) = \int_L f(x_1,x_2)\,\mathrm{d}x \tag{1}$$

Concretely, the parametrization of any straight line L with respect to arc length t can be expressed as $(x_1(t),x_2(t)) = ((t\sin\alpha + s\cos\alpha),(-t\cos\alpha + s\sin\alpha))$, where s represents the distance of L from the origin, and α is the angle the normal vector to L makes with the x_1-axis. Consequently, the Radon transform can be defined as $R(s,\alpha) = \int_L f(x_1,x_2),\mathrm{d}l$. The distinction between the internal problem and a full-scan CT primarily resides in the range of s. As depicted in figure 1, for the internal problem, $s \in [-r,r]$, whereas for a typical full-scan CT, the data collection extends over $s \in [-R,R]$.

According to HOT theory (Zhao *et al* 2015), suppose that $f(x)$ is a piecewise nth order polynomial function in the ROI (Ω_r); that is, Ω_r can be decomposed into finitely many sub-domains $\{\Omega\}_{i=1}^m$, such that

$$f(x) = f_i(x), x \in \Omega_i, 1 \leqslant i \leqslant m \tag{2}$$

where $f_i(x)$ is an nth order polynomial function, and sub-domains satisfy $\Omega_i \cap \Omega_j = \varnothing, i \neq j, \cup\Omega_{i=1}^m = \Omega_r$, and each sub-domain Ω_i is adjacent to its neighboring sub-domain Ω_j with piecewise smooth boundaries $\Gamma_{i,j}$. For any candidate image f, define the n+1th order total variation (TV) as

$$\mathrm{HOT}_{n+1}(f) = \sum_{i=1}^{m} \int_{\Omega_i} \sqrt{\sum_{k=0}^{n+1} \left(\frac{\partial^{n+1} f}{\partial x_1^k \partial x_2^{n+1-k}} \right)^2} \tag{3}$$

$$f(x) = \arg\min \mathrm{HOT}_{n+1}(f) \tag{4}$$

Here, HOT is a simplified version. The general definition of HOT can be found in Yang *et al* (2010, 2012b), and when $n = 0$, the $(n+1)$th order TV reduces to the classical TV.

Specifically, considering the second-order case (Yang *et al* 2010), suppose an object image is a piecewise linear function in Ω_r; that is, Ω_r can be decomposed into finitely many sub-domains: $\{\Omega_{i=1}^m\}$, such that

$$f_i(x) = a_i x_1 + b_i x_2 + c_i, x \in \Omega_i, 1 \leqslant i \leqslant m \tag{5}$$

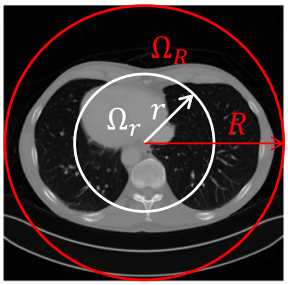


Figure 1. Schematic comparison between full-scan CT and a region-of-interest (ROI) for interior tomography. In full-scan CT, the entire scanning area Ω_R (radius R) is irradiated, covering both target region and surrounding tissuses. The ROI scan restricts irradiation to the smaller area $\Omega_r (r < R)$, focusing radiation dose on the diagnostically critical region.

Algorithm 1. SART+HOT algorithm.

1: Input: Initial value $f^0(x)$,truncated projection data p
2: Output: $f^n(x)$
3: **for** each $i \in [1, N]$ **do**
4: $\quad f^{i+\frac{1}{2}} = SART(f^i(x), p)$
5: $\quad f^{i+1} = arg\min_f HOT(f)$
6: **end for**

and each sub-domains Ω_i is adjacent to its neighboring sub-domains Ω_j with piecewise smooth boundaries Γ_i, j. Then

$$\begin{aligned} \mathrm{HOT}_2(f) = &\sum_{i=1}^{m}\sum_{j>i}\int_{\Gamma_{i,j}} |\left(a_i - a_j\right)x_1 + \left(b_i - b_j\right)x_2 + \left(c_i - c_j\right)|\mathrm{d}s \\ &+ \int_{\Omega_r} \min\left\{ \sqrt{\frac{1}{4}\left(\frac{\partial^2 f}{\partial x_1^2} - \frac{\partial^2 f}{\partial x_2^2}\right)^2 + \left(\frac{\partial^2 f}{\partial x_1 \partial x_2}\right)^2} \right. \\ &\left. + \frac{1}{2}\left|\frac{\partial^2 f}{\partial x_1^2} + \frac{\partial^2 f}{\partial x_2^2}\right|, \sqrt{\left(\frac{\partial f}{\partial x_1}\right)^2 + \left(\frac{\partial f}{\partial x_2}\right)^2} \right\} \mathrm{d}x_1 \mathrm{d}x_2 \end{aligned} \tag{6}$$

The complete reconstruction process, when HOT is applied to the internal problem, can be conceptualized as the following optimization problem:

$$f(x) = \arg\min_f \frac{\mu}{2}\|Rf(x) - p\|_w^2 + \mathrm{HOT}(f) \tag{7}$$

where p represents the collected truncated projection data. To address this issue, the approach involves estimating the data consistency term and alternately minimizing the HOT until a high-quality image is obtained. The alternating iterative reconstruction process is shown in algorithm 1.

3. Methods

We hereby introduce how the Fourier-improved high-order TV network can address the interior tomography, detailing each segment's contributions.

3.1. Whole architecture: Fourier-enhanced HOT iteration network

First, we briefly introduce the overall architecture of our proposed network model. As illustrated in figure 2, the network undergoes an iterative process. Each iteration module comprises two primary segments. The first segment, the coarse segment, is focused on feature extraction and embedding based on high-order TV. The second segment, the fine segment, is dedicated to enhancing high-frequency information through Fourier transform, thereby improving the algorithm's accuracy. Let N denote the desired number of iterations, and $\theta^k (k = 1, \cdots, N)$ represent the learnable weight parameters of $PD, \mathcal{H}_1, \mathcal{H}_2, \mathcal{K}$. Consequently,

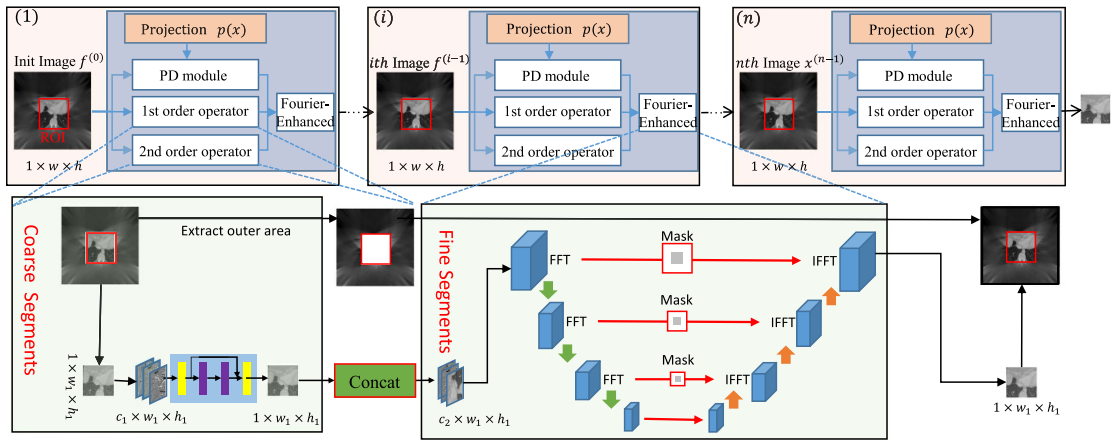


Figure 2. Illustration of the Fourier-enhanced high order TV iteration network. The top row illustrates the processing of iterations, while the bottom rows present two detailed schematics: one depicting the fusion of image gradients, and the other showing the U-Net model with enhanced frequency domain details post-Fourier transform. Notably, c_1 represents the input channels corresponding to different orders of derivatives $c_1 = 2$ for the first order and $c_1 = 3$ for the second order. Similarly, c_2 represents the input channels of the fine segment, with $c_2 = k + 1$ for the kth order TV.

Algorithm 2. FeHOT algorithm (coarse-scale).

Input: Initial value $f^0(x)$,truncated projection data p
Output: $f^n(x)$
for each $i \in [1, N]$ **do**
  # HOT-based network for ROI reconstruction
  # data consistency item
  $f^{i+\frac{1}{2}} \leftarrow PD(f^i, p; \theta^i_{PD})$
  # the first order gradients items embedding
  $\delta f^i \leftarrow \mathcal{H}_1(f^i_{x_1}, f^i_{x_2}; \theta^i_{\mathcal{H}_1})$
  # the second order gradient items embedding
  # $\delta^2 f^i \leftarrow \mathcal{H}_2(f^i_{x_1x_1}, f^i_{x_1x_2}, f^i_{x_2x_2}; \theta^i_{\mathcal{H}_2})$
  # Fourier-enhanced module
  $f^{i+1} \leftarrow \mathcal{K}(f^{i+\frac{1}{2}}, \delta f^i, \delta^2 f^i; \theta^i_{\mathcal{K}})$
end for

the FeHOT algorithm can be succinctly summarized at a coarse scale, as presented in algorithm 2. Thus, FeHOT consists of N blocks that share the same network structure but possess distinct network parameters.

3.2. Improved HOT iteration network

The objective is to develop a learned HOT method for the interior tomography. We propose a parameterized plug-and-play HOT. Specifically, we modify the HOT iterations in algorithm 1 as follows:

1) In the data consistency item, we adopt the learned primal-dual algorithm (PD-Net)(Adler and Öktem 2018) to replace the SART algorithm in algorithm 1. Although the implementation of the SART operator is feasible according to Ma *et al* (2021), the non-parallelism of its forward and reverse processes, calculated angle-by-angle, prolongs the training process. In Ma *et al* (2021), the training time for the reconstruction process is reduced through a transition from partial to overall training, whereas PD-Net more conveniently facilitates the direct transformation of end-to-end projection to image.

2) In the HOT item, as per equation(6), HOT comprises two terms: one is the boundary term, which pertains to certain details of the reconstruction part at the boundary that can be easily learned by the network. The other is to optimize the internal structure of the region using its partial derivatives, where the discretized format of partial derivatives can be implemented using 3×3 convolution kernels. Further

specifics are detailed below:

$$\frac{\partial f}{\partial x_1} = f \otimes \begin{pmatrix} 0 & 0 & 0 \\ \frac{1}{2} & 0 & -\frac{1}{2} \\ 0 & 0 & 0 \end{pmatrix}, \frac{\partial f}{\partial x_2} = f \otimes \begin{pmatrix} 0 & \frac{1}{2} & 0 \\ 0 & 0 & 0 \\ 0 & -\frac{1}{2} & 0 \end{pmatrix},$$
$$\frac{\partial^2 f}{\partial x_1^2} = f \otimes \begin{pmatrix} 0 & 0 & 0 \\ \frac{1}{2} & -1 & \frac{1}{2} \\ 0 & 0 & 0 \end{pmatrix}, \frac{\partial^2 f}{\partial x_2^2} = f \otimes \begin{pmatrix} 0 & \frac{1}{2} & 0 \\ 0 & -1 & 0 \\ 0 & \frac{1}{2} & 0 \end{pmatrix},$$
$$\frac{\partial^2 f}{\partial x_1 \partial x_2} = f \otimes \begin{pmatrix} \frac{1}{4} & 0 & -\frac{1}{4} \\ 0 & 0 & 0 \\ -\frac{1}{4} & 0 & \frac{1}{4} \end{pmatrix} \tag{8}$$

Then the simplest learned HOT optimization method is as follow:

$$\delta^2 f \leftarrow \mathcal{H}_2 \left(\sqrt{\frac{1}{4}\left(\frac{\partial^2 f}{\partial x_1^2} - \frac{\partial^2 f}{\partial_2^2 x_2}\right)^2 + \left(\frac{\partial^2 f}{\partial x_1 \partial x_2}\right)^2} + \frac{1}{2}\left|\frac{\partial^2 f}{\partial x_1^2} + \frac{\partial^2 f}{\partial x_2^2}\right|; \theta_{\mathcal{H}_2} \right)$$
$$\delta f \leftarrow \mathcal{H}_1 \left(\sqrt{\left(\frac{\partial f}{\partial x_1}\right)^2 + \left(\frac{\partial f}{\partial x_2}\right)^2}; \theta_{\mathcal{H}_1} \right) \tag{9}$$

However, this performance did not significantly improve compared with traditional HOT algorithms.

To achieve substantial improvements, guided by Adler and Öktem (2018), the following modifications were made to the improved HOT algorithm.

- Instead of using explicit expressions, these were transformed into implicit and flexible learnable parameters. Although the expression remains consistent in each iteration, different learned parameters are utilized in operations to enhance the network's performance and the accuracy of the reconstructed images.
- To render our network as reasonable as possible, the activation functions of the three convolutional layers used in the network for each order of TV are square activation, absolute activation, and square root activation. This choice is due to the sequential appearance of the square root term, absolute value term, and square term in explicit expressions.

Thus, the improved HOT algorithm is as follows:

$$\delta^2 f \leftarrow \mathcal{H}_2 \left(\frac{\partial^2 f}{\partial x_1^2}, \frac{\partial^2 f}{\partial x_2^2}, \frac{\partial^2 f}{\partial x_1 \partial x_2}; \theta_{\mathcal{H}_2} \right)$$
$$\delta f \leftarrow \mathcal{H}_1 \left(\frac{\partial f}{\partial x_1}, \frac{\partial f}{\partial x_2}; \theta_{\mathcal{H}_1} \right) \tag{10}$$

Let $\mathcal{C}$ denote the convolution operation, while $\mathcal{S}, \mathcal{A}$, and $\mathcal{R}$ represent the square activation, absolute activation, and square root activation functions, respectively. Accordingly, $\mathcal{H}_1$ and $\mathcal{H}_2$ are defined using the following identical operators:

$$\mathcal{H} = \mathcal{R} \circ \mathcal{C} \circ \mathcal{A} \circ \mathcal{C} \circ \mathcal{S} \circ \mathcal{C}. \tag{11}$$

where the symbol $\circ$ denotes the composition of two operators in the usual sense. The distinction between $\mathcal{H}_1$ and $\mathcal{H}_2$ lies in the number of input channels; $\mathcal{H}_1$ utilizes two channels, whereas $\mathcal{H}_2$ uses three channels, as illustrated in figure 2.

3.3. Fourier-enhanced module for high-frequency information fusing

Based on the aforementioned data and high-order TV items, we concatenate these components and input them into the traditional U-Net model for iterative resolution, as detailed in the evaluation and ablation study sections. However, its performance remains constrained by the assumption of a piecewise polynomial function. Inspired by Wu *et al* (2023), the adaptive wavelet sub-network effectively reduces disturbances in the training data and enhances score-based generative training stability. Similarly, we observe that high-frequency information, which is often lost in the detailed image structures through traditional methods, is challenging for regularization methods to preserve. A potential solution involves extracting this information and integrating it into our network to enhance the accuracy of reconstructed images. Initially, we implement Fourier transform on the input feature tensor in the skip connection of the U-Net and employ

spectral modulation in the Fourier domain to selectively reduce low-frequency components of the skip features (Si *et al* 2023).

Typically, for the kth block of the U-Net decoder module, as illustrated in the fine segment of figure 2, let e_k denote the feature map propagated through the corresponding skip connection. This operation is performed as follows:

$$\begin{aligned} \mathcal{F} &= fft(e_k) \\ \mathcal{F}' &= \mathcal{F} \cdot M_k \\ d_k &= ifft(F') \end{aligned} \tag{12}$$

where *fft* and *ifft* denote the Fourier/inverse Fourier transform, respectively. The symbol · represents element-wise multiplication. The term d_k refers to the enhanced skip feature map, which is concatenated with the decoded feature map for the subsequent layers in the U-Net architecture. The term M_k is a mask designed to function based on the scaling of the Fourier coefficients, serving to implement the frequency-dependent scaling factor s_k and the frequency factor $r_k \in [0, 1]$:

$$M_k = \begin{cases} s_k & r < r_k \\ 1 & \text{otherwise} \end{cases} \tag{13}$$

where r is normalized radial distance from image center. When $s_k = 1$, the Fourier-enhanced module effectively functions as the classical U-Net. In our experiments, we set the scale factor $s_k = 0.5$ and the frequency factor $r_k = 0.5$.

4. Experiments

We conducted several experiments and analyzed reconstructed images to evaluate the performance of the proposed FeHOT.

4.1. Dataset and settings

To evaluate the imaging performance of FeHOT under conditions that are as realistic as possible, we employ two different types of datasets. The first is the dataset provided by the 2021 AAPM challenge (The DL-sparse-view CT challenge), referred to as the AAPM dataset, which adheres to the piecewise function assumption (Sidky and Pan 2022). The second dataset comprises clinical data from human abdomen, chest, and lung, obtained from the Cancer Genome Atlas Esophageal Carcinoma (TCGA-ESCA)[4](Lucchesi and Aredes 2016), referred to as the medical dataset. A typical slice from each dataset is displayed in figure 3. Both noisy and noise-free projection data are simulated, with Poisson noise, specified by the photon intensity, denoted as I_0, invariably added to the raw data, as follows:

$$p_{\text{noisy}} = -\ln\left(\frac{\text{Poisson}\left(I_0 \times \mathrm{e}^{-p}\right)}{I_0}\right)$$

where the photon intensity I_0 is set at 5×10^4.

- **AAPM dataset.** Following the settings of the AAPM dataset, this dataset contains 4000 image pairs derived from a 2D breast CT simulation. Ground truth images are used to perform ROI simulation and validate the proposed FeHOT algorithm. Specifically, 3500 images, each with dimensions of 512×512 pixels, are used for training, and 500 images are reserved for testing. The geometrical parameters are also listed in table 1.
- **Medical dataset.** This dataset comprises data from 50 patient volumes, totaling 5890 image slices, each with a resolution of 512×512 pixels. The training dataset includes randomly selected volumes from 40 patients, totaling 4086 slices, while the remaining 10 patients' volumes, comprising 804 slices, are reserved for evaluation. A two-dimensional fan-beam geometry is adopted for the virtual CT system, and the geometrical parameters are also listed in table 1.

[4] TCGA-ESCA:www.cancerimagingarchive.net/.

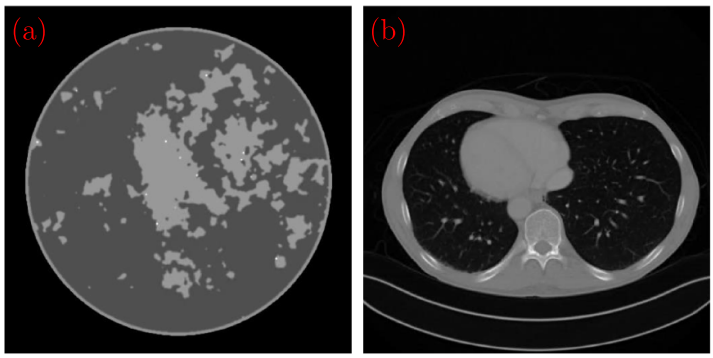


Figure 3. One typical slice of two dataset, (a) The DL-sparse-view CT chanllenge dataset, (b) TCGA-ESCA dataset.

Table 1. Geometrical scanning parameters of the AAPM dataset and medical dataset.

	AAPM data	Medical data
Number of detector units	300	600
Scanning angle (deg)	$[0^\circ, 360^\circ]$	$[0^\circ, 360^\circ]$
X-ray source to the rotation center (mm)	500	500
X-ray source to detector distance (mm)	1000	1000
Scanning angular interval (deg)	1	1
Detector uint width(mm)	1.1	0.55
Field of view(pixel)	112	80
Pixel size(mm)	0.7268	1.0175
Reconstructed image size(pixel)	512×512	512×512

4.2. Implementation details

The proposed FeHOT is implemented in Python using the PyTorch framework. All experiments were conducted on a PC equipped with an Intel(R) Xeon(R) Platinum 8358 CPU @ 2.60GHz and 384GB RAM. The training stage was executed on a single NVIDIA GeForce RTX 4090 graphics processor with 24 GB video memory.

For the Radon transform, we utilized the *RayTransform* operator and its adjoint from the Operator Discretization Library (ODL)[5] (Adler and Öktem 2018), employing the GPU-accelerated *astra_gpu* backend (Van Aarle *et al* 2015, 2016). To integrate the Radon Transform as a layer within PyTorch, we invoked the *OperatorModule* functionality of ODL.

To optimize performance, the network's hyperparameters, such as the learning rate and size of the convolution kernels, were carefully calibrated. Specifically, the number of base features in the U-Net was set at 32, using 3×3-sized convolutional filters and rectified linear unit operations (ReLU). The MSE loss function was employed, minimized using the adaptive moment estimation (Adam) optimizer. The training samples were randomly shuffled, and the initial learning rate was set at 1×10^{-4}, adjusted using a standard exponential decay scheduling technique. Furthermore, the entire network was manually set to iterate five times, with each training data utilized 100 times in training for the medical dataset, resulting in $epoch = 100$ and a batch size of 16. For the AAPM dataset, $batch_size = 8$ and $epoch = 300$. The training process took approximately 15 h for the medical data and 71 h for the AAPM dataset.

4.3. Competing methods

For a fair comparison with classical analytical, iterative algorithms, and deep learning algorithms, we employed standard FBP, HOT and two state-of-the-art learning-based reconstruction methods: Attention gated network (AG-Net) (Schlemper *et al* 2019, Oktay 2022) and the learned PD-Net. These methods are briefly summarized here, with full descriptions available in the cited references.

FBP reconstruction is a typical analytical CT image reconstruction method implemented with the *ASTRA Toolbox*[6]. Specifically, the FBP algorithm utilizes a Shepp–Logan filter and is executed using the *FBP_CUDA* function in the ASTRA Toolbox. For the HOT method, the code involves custom CUDA kernels

[5] ODL: https://github.com/odlgroup/odl.

[6] ASTRA Toolbox: https://astra-toolbox.com/.

manually written for SART reconstruction, combined with MATLAB code for HOT constraints. The regularization parameter μ is selected to maximize the peak signal to noise ratio (PSNR), and 300 iterations are set as an empirical stopping criterion.

For the deep learning methods, the hyperparameters are almost identical to those of the proposed method, such as epochs, learning rate, batch size, and loss function. However, certain differences are noted: Specifically, the AG-Net[7] functions as a post-processing step, where the input images are reconstructed images from the FBP algorithm. For PD-Net, we implemented its interface using open-source code of PD-Net[8] and the ODL library. Due to the slow convergence of PD-Net during training, epochs are set at 400 and the number of iterations is adjusted to 10 for a fairer comparison. The PD-Net architecture employs 10 primal and 10 dual operators.

4.4. Experimental results

In this sub-section, the proposed FeHOT will be compared with classical analytical, iterative algorithms, and deep learning methods.

4.4.1. AAPM dataset

Firstly, we conduct experimental analysis on the AAPM dataset. Visually, as observed in figure 4, for the noise-free case, ROI can be effectively reconstructed by all competing methods and the proposed method, except for the FBP algorithm. However, notable differences between the methods exist. Specifically, the reconstruction results of HOT and PD-Net exhibit a tendency to slightly weaken the edges of the reconstructed image and excessively smoothen the high attenuation areas. Meanwhile, the proposed method, FeHOT($n = 2$) , exhibits edge effects in the reconstructed image, characterized by a bright edge near the boundary, which does not appear when $n = 1$. This phenomenon may result from the reconstructed image adhering to the piecewise constant assumption and the forced use of second-order gradients, leading to edge continuity and excessive smoothness. In the subsequent study of medical data, when the reconstructed image does not satisfy the piecewise constant assumption, this phenomenon is absent in FeHOT($n = 2$) reconstructed images.

Table 2 lists the statistical results of different algorithms in terms of PSNR , structural similarity index measure (SSIM), and root mean square error(RMSE) on the AAPM dataset. From this perspective, the various performance evaluation indicators corroborate our visual analysis and demonstrate the promising performance of our proposed method in addressing the interior tomography, as indicated by the competitive PSNR, SSIM, and RMSE results. Meanwhile, FeHOT($n = 1$) significantly outperforms other methods across all metrics (PSNR = 55.039, SSIM = 0.9998, RMSE = 0.0005), demonstrating its strong adaptability to piecewise constant assumptions in AAPM data, where FeHOT($n = 2$) suffers from edge oversmoothing due to enforced second-order gradients (visible as bright edges in the residual map of figure 4(g)), yet still surpasses traditional methods (e.g. HOT, PD-Net).

4.4.2. Medical dataset

Figure 5 shows the reconstructions for a representative medical slice. The first two rows display noise-free results, while the last two rows show the noisy results. The results, starting from the second column, are reconstructed by FBP, HOT, AG-Net, PD-Net, FeHOT($n = 1$) and FeHOT($n = 2$). First, we focus on the noise-free results. The traditional FBP algorithm, while producing relatively clear object details, introduces significant cupping artifacts in the reconstructed image due to out-of-FOV (field-of-view) imaging. The iterative HOT algorithm mitigates these cupping artifacts; however, its assumptions of piecewise constancy or smoothness may severely distort weak edges and structures. More advanced methods such as the post-processed AG-Network and the iterative PD-Net demonstrate superior performance over traditional algorithms. Although AG-Net tends to exhibit overfitting, leading to unique artifacts in the reconstructed images, PD-Net maintains higher image quality by alternating between data consistency and regularization terms. Nevertheless, PD-Net still fails to fully recover details and preserve edges due to unaccounted propagation of high-frequency components.

Next, we analyze the reconstructed results of noisy data. Overall, the results are similar to the noise-free cases, with minor differences. Specifically, the noise in the FBP reconstruction results obscures certain detailed structures. Block effect and salt-and-pepper-like noise are exacerbated in the HOT method's

[7] AG-Net code: https://github.com/ozan-oktay/Attention-Gated-Networks.
[8] PD-Net code: https://github.com/adler-j/learned_primal_dual.

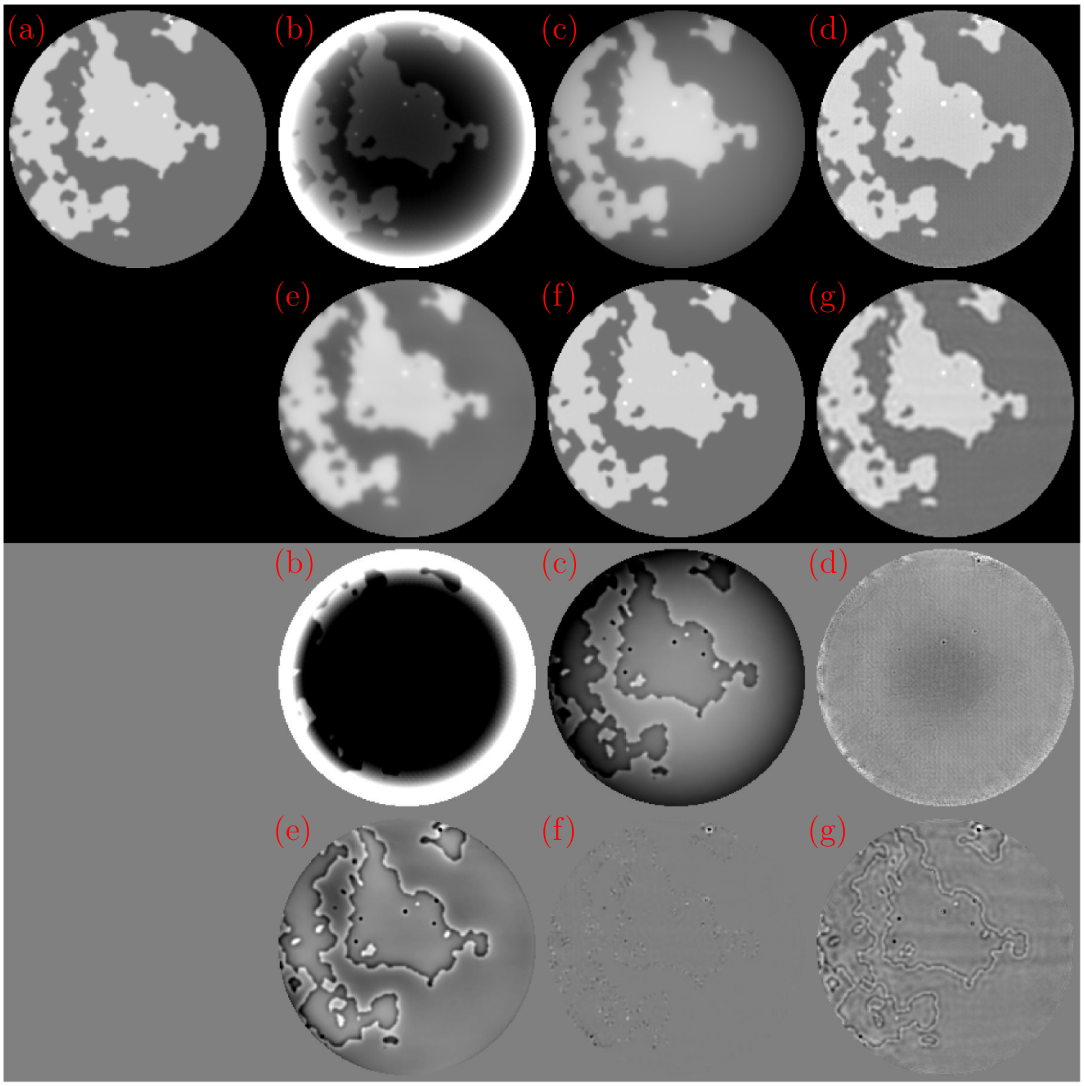


Figure 4. Reconstructions of a sample test slice in the AAPM dataset are shown. The first two rows display the results of the noise-free case, and the last two rows show the difference images between the corresponding reconstruction methods and the ground truth. (a) Ground truth, (b) FBP method with display windows $[-600, -100]$, (c) HOT algorithm ($\mu = 300$) with display windows $[-150\text{HU}, 400\text{HU}]$, (d) AG-Net, (e) PD-Net, (f) FeHOT($n = 1$), (g) FeHOT($n = 2$). The display windows of (d)–(g) are set to $[-180\text{HU}, 360\text{HU}]$ for the reconstructed images, while all difference images are set to $[-100\text{HU}, 100\text{HU}]$.

reconstructed image. Meanwhile, the proposed methods demonstrate smoothness and a closer approximation to the ground truth.

Focusing on the difference images, displayed in figure 6, our proposed method demonstrates significant advantages in edge preservation, particularly in the noise-free case. This further corroborates our analysis from the reconstructed images. Specifically, PD-Net and AG-Net, while outperforming traditional methods, exhibit notable edge errors in residual maps (figures 6(d) and (e)) due to high-frequency information loss. Although FeHOT($n = 2$) exhibits a subtle visual advantage over $n = 1$ in medical data, both methods significantly outperform traditional methods. Notably, FeHOT($n = 2$) avoids the edge effects observed in the AAPM dataset (figure 4(g)).

To quantitatively compare the results, we computed the quantitative results for PSNR, SSIM, and RMSE, as shown in table 3. The proposed algorithm achieves an improved approximation to the ground truth image, indicated by the highest PSNR, SSIM, and lowest RMSE. This solid evidence strongly aligns with the visual observations. In general, For noise-free scenario, FeHOT achieves marginally higher PSNR, SSIM, indicating superior detail recovery for complex anatomical structures, while the performance of $n = 1$ and $n = 2$ is basically the same. For noisy scenario, FeHOT($n = 2$) excels with higher PSNR (39.241) and SSIM (0.9899), demonstrating the robustness of its second-order TV regularization against noise, and substantially better than FeHOT($n = 1$) for objects that do not satisfy the piecewise constant function assumption.

Table 2. Statistical results of different algorithms in terms of PSNR, SSIM, and RMSE on the AAPM dataset.

	PSNR	SSIM	RMSE
FBP	4.116	0.7639	0.1881
HOT	27.903	0.9958	0.0122
AG-Net	28.492	0.9972	0.0114
PD-Net	38.918	0.9950	0.0034
FeHOT($n = 1$)	**55.039**	**0.9998**	**0.0005**
FeHOT($n = 2$)	44.477	0.9978	0.0018

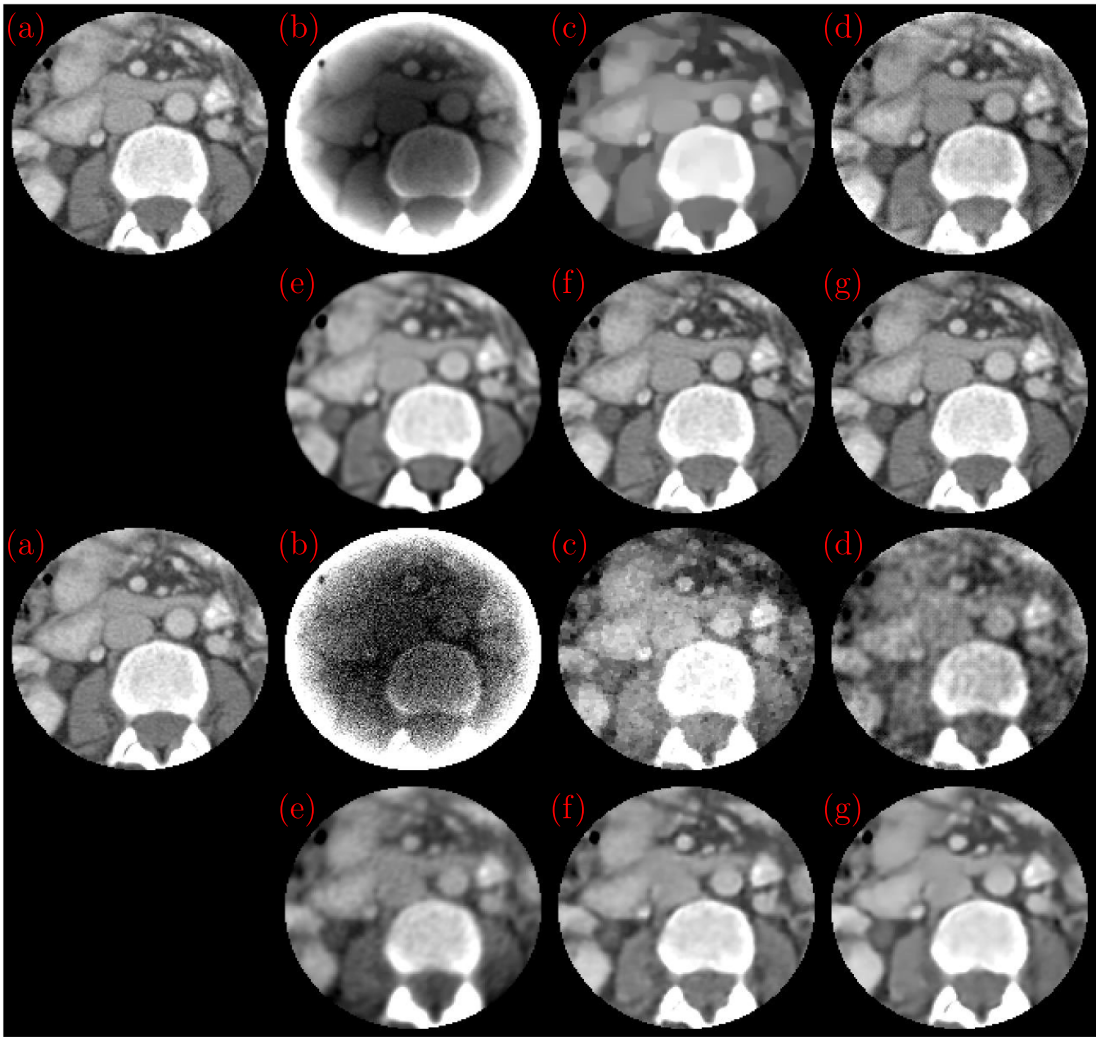


Figure 5. Reconstructions of a sample test slice in the medical dataset are shown. The first two rows display the results of the noise-free case, while the last two rows show the results of the noisy-data case. (a) Ground truth, (b) FBP method with display windows [−600, −100], (c) HOT algorithm ($\mu = 200$ for noise-free and $\mu = 150$ for noisy-data) with display windows [−150HU, 400HU], (d) AG-Net, (e) PD-Net, (f) FeHOT($n = 1$), (g) FeHOT($n = 2$). The display windows for (d)–(g) are set to [−180HU, 360HU].

4.5. Ablation study

4.5.1. Necessity of frequency-domain reweighting

To verify the performance of Fourier-enhanced reconstruction and analyze it effectively, we set different frequency-dependent scaling factors s_k for experimental analysis. Note that $s_k = 1$ can be considered as no Fourier enhancement. As shown in figure 7, the structure of the scanned object can be effectively reconstructed by FeHOT with different s_k values and order factors n. However, certain notable differences exist, primarily due to s_k. Specifically, in the noise-free case, compared with the frequency-dependent scaling factor introduced in the skip connection, the images reconstructed by FeHOT exhibit relative smoothness in the detailed structure. The detailed structure can be better recovered by reducing the weight of low frequencies in the Fourier domain, thus enhancing the transmission of high-frequency signals. Additionally,

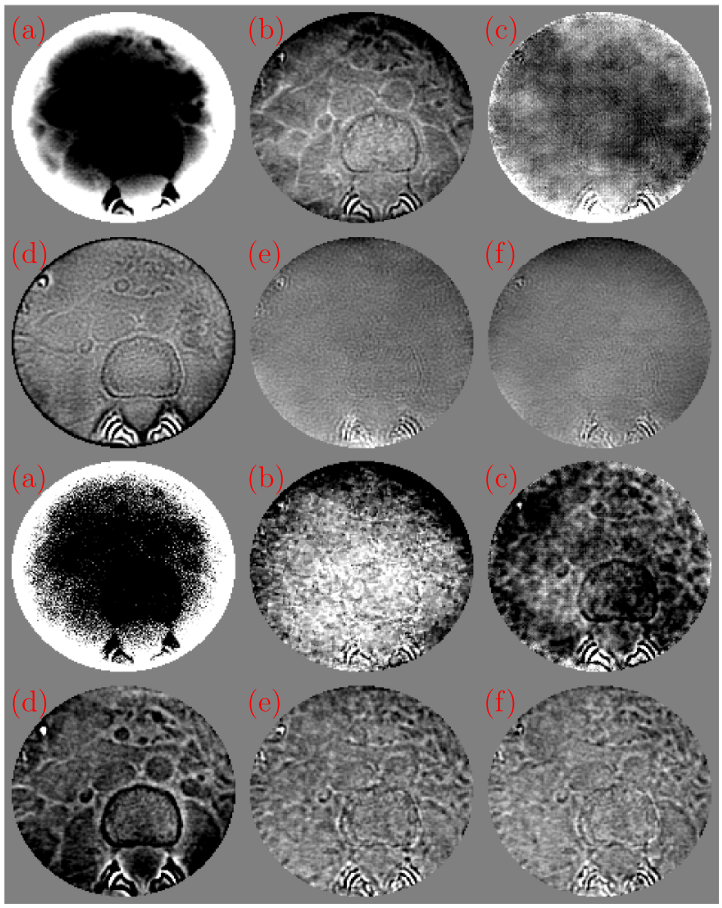


Figure 6. The difference images in figure 5 between the corresponding reconstruction methods and the ground truth are shown. (a) FBP, (b) HOT, (c) AG-Net, (d) PD-Net, (e) FeHOT($n = 1$), (f) FeHOT($n = 2$). The display windows are set to [−100HU, 100HU].

Table 3. Statistical results of different algorithms in terms of PSNR, SSIM, and RMSE on the medical dataset.

		PSNR	SSIM	RMSE
Noise-free	FBP	6.514	0.7243	0.2029
	HOT	32.530	0.9901	0.0101
	AG-Net	33.408	0.9888	0.0092
	PD-Net	33.425	0.9841	0.0092
	FeHOT($n = 1$)	**41.173**	0.9976	**0.0037**
	FeHOT($n = 2$)	40.998	**0.9981**	0.0038
Noisy data	FBP	6.457	0.5019	0.2043
	HOT	31.390	0.9712	0.0116
	AG-Net	31.313	0.9732	0.0117
	PD-Net	31.078	0.9759	0.0120
	FeHOT($n = 1$)	37.967	0.9876	0.0054
	FeHOT($n = 2$)	**39.241**	**0.9899**	**0.0047**

observing the difference images in figure 7, Fourier enhancement is found to be more advantageous for preserving edge structures. This experimental result strongly supports the necessity of the fine reconstruction process.

In the presence of noise, due to its influence, the reconstructed results under different parameters exhibit smoothness in details, rendering it difficult to distinguish the reconstruction quality visually. Table 4 shows the performance evaluation indicators, reflecting the imaging quality under different parameters. By comparing these indicators, the proposed Fourier-enhanced method provides substantial improvemence gains. For noise-free,PSNR of FeHOT($n = 1, s_k = 0.5$) (41.173) shows a 13.5% improvement over $s_k = 1.0$ (36.301), where, for noisy cose, RMSE of FeHOT($n = 2, s_k = 0.5$) (0.0047) is 38% lower than $s_k = 1.0$

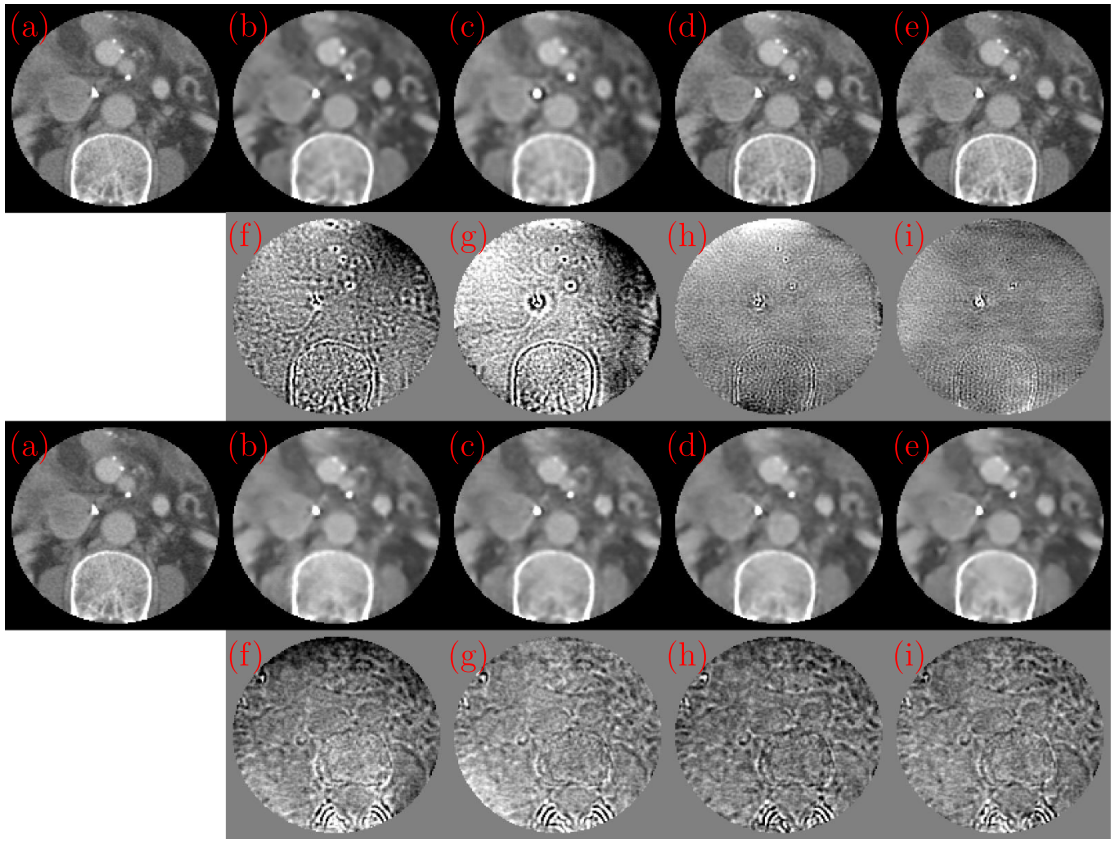


Figure 7. The reconstructed and difference images with different hyperparameters of frequency-dependent scaling factors s_k. The first and third rows show the reconstructed images for noise-free and noisy data cases. (a) Ground truth, (b) FeHOT($n = 1$,$s_k = 1.0$), (c) FeHOT($n = 2$,$s_k = 1.0$), (d) FeHOT($n = 1$,$s_k = 0.5$), (e) FeHOT($n = 2$,$s_k = 0.5$). The display windows of the reconstructed images are set to [-180HU, 360HU]. The even rows display the difference images with display windows set to [-50HU, 50HU] for noise-free cases and [-100HU, 100HU] for noisy-data cases.

Table 4. Quantitative results with different frequency-dependent scaling factors s_k for the medical dataset.

	FeHOT(n, s_k)	PSNR	SSIM	RMSE
Noise-free	$n = 1$, $s_k = 1.0$	36.301	0.9877	0.0066
	$n = 2$, $s_k = 1.0$	37.011	0.9875	0.0061
	$n = 1$, $s_k = 0.5$	**41.173**	0.9976	**0.0037**
	$n = 2$, $s_k = 0.5$	40.998	**0.9980**	0.0038
Noisy data	$n = 1$, $s_k = 1.0$	35.994	0.9847	0.0068
	$n = 2$, $s_k = 1.0$	34.990	0.9842	0.0076
	$n = 1$, $s_k = 0.5$	37.967	0.9876	0.0054
	$n = 2$, $s_k = 0.5$	**39.241**	**0.9899**	**0.0047**

(0.0076). $s_k = 0.5$ suppresses low-frequency dominance and enhances high-frequency features, preserving edge details with smaller residuals (figures 7(d) and (e)).

4.5.2. Real experiments

Real experiments were carried out on the head of a sheep to study the robustness and well-posedness of our proposed method in practical applications. The scanning configuration of the fan-beam data was set as follows. We used a linear detector with 2048 detector units and a 0.2 mm unit pitch. The SOD and SDD were 415.0 and 781.0 mm, respectively. A total of 1080 projections were scanned per rotation. The obtained fan-beam projections were interpolated to adapt to the noise-free trained network models for testing, and the corresponding reconstruction results are illustrated in figure 8.

The experimental results on real-data reconstruction demonstrate the practical effectiveness of the proposed FeHOT method. Compared to post-processing networks (e.g. AG-Net), which suffer from overfitting and persistent artifacts in cross-data scenarios, and iterative networks (e.g. PD-Net) that compromise high-frequency details for overly smooth outputs, FeHOT achieves superior balance between

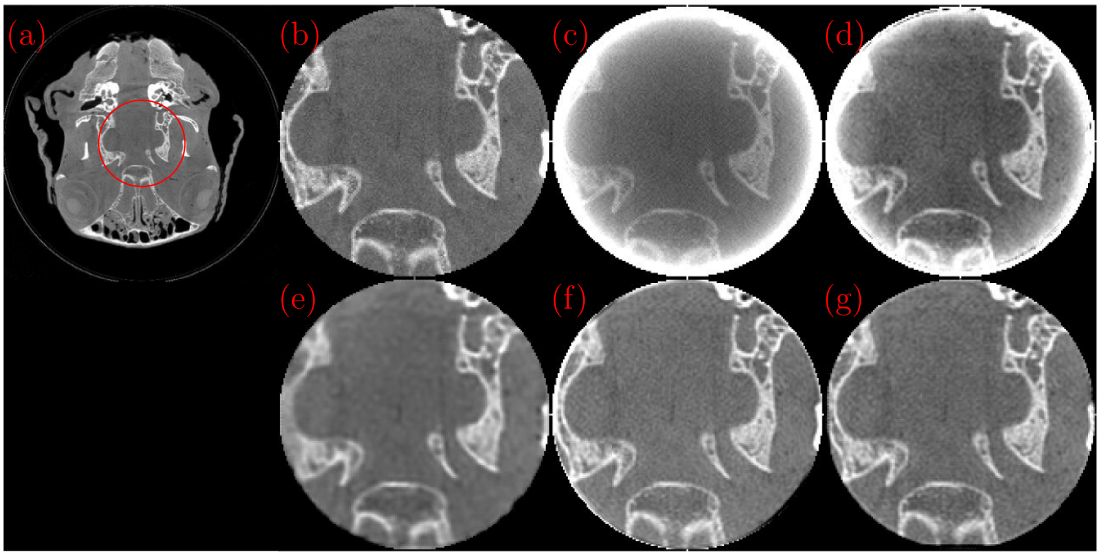


Figure 8. Reconstructions of a sample test slice in the real data are shown. (a) FBP for full-scan data, (b) The zoom-in of the area marked with a red circle in (a), (c) FBP method for ROI with display windows $[0.025, 0.6]\,\mathrm{mm}^{-1}$, (d) AG-Net, (e) PD-Net, (f) FeHOT($n = 1$), (g) FeHOT($n = 2$). The display windows for (a), (b), (d)–(e) are set to $[0.05, 0.3]\,\mathrm{mm}^{-1}$.

artifact suppression and structural fidelity. Specifically, FeHOT($n = 2$) exhibits enhanced CT value accuracy over FeHOT($n = 1$) in localized regions of interest, validating its parameter-driven robustness. These findings highlight FeHOT's adaptability to complex real-world data, addressing limitations of existing methods in generalizing beyond simulated environments while maintaining high-resolution detail preservation.

5. Discussions and conclusion

This study presents a novel iterative network-based HOT method for the interior tomography, capable of reweighting high and low frequencies in the Fourier domain. The proposed FeHOT enables purely data-driven supervised learning for the interior tomography using two-dimensional projection data. The entire process consists of two strategies:

- Coarse reconstruction process: FeHOT employs the unrolling technique based on HOT theory.
- Fine reconstruction process: Introducing frequency-dependent scaling factors and frequency factors in the skip connections of the U-Net effectively overcomes the restrictions imposed by the piecewise constant or polynomial assumptions usually made for CT images.

The proposed method achieves promising results. While existing methods face critical limitations-FBP's truncation artifacts, HOT's over-smoothing, AG-Net's dependency on flawed inputs, and PD-Net's high-frequency structural information is not well–FeHOT addresses these gaps through a novel integration of unrolled HOT regularization and frequency-domain enhancement. By attenuating low-frequency skip features in U-Net, FeHOT uniquely preserves high-frequency details (e.g. edges, textures) that traditional MBIR and post-processing CNNs fail to recover. This is validated quantitatively (tables 2 and 3: FeHOT achieves >40dB PSNR vs. 33 dB for PD-Net/AG-Net) and qualitatively (figures 5 and 6: reduced cupping artifacts and sharper structures). Clinically, FeHOT's efficiency (5 iterations) and robustness to noise make it a practical solution for low-dose ROI imaging, with potential extensions to dynamic CT or multi-modal reconstruction.

Although the proposed FeHOT achieves state-of-the-art reconstruction quality for the interior tomography, it also has several limitations that should be discussed. For instance, as shown in figure 5, certain local structures are lost in the reconstructed image of noisy data, rendering it difficult to ascertain whether this is caused by noise. Moreover, incorporating Fourier transform in the skip connections into the traditional U-Net backbone inevitably results in a slight increase in space and computational costs. The manual tuning of the scale and frequency factors in the proposed method may indeed be suboptimal, and exploring optimal parameter configurations is a key focus of future study. Network-driven parameter optimization will serve as the primary direction to address this challenge.

In conclusion, motivated by pioneering results of HOT, we have developed a state-of-the-art iterative framework that extends HOT for the interior tomography into a deep-learning network, called FeHOT. As verified in comprehensive experiments and ablation studies, the proposed method effectively restores fine textures and edges in the reconstructed images.

Data availability statement

The data that support the findings of this study are openly available at the following URL/DOI: www.aapm.org/GrandChallenge/DL-sparse-view-CT/.

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